



TerraScope: Pixel-Grounded Visual Reasoning for Earth Observation

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<https://shuyansy.github.io/terrascope/>

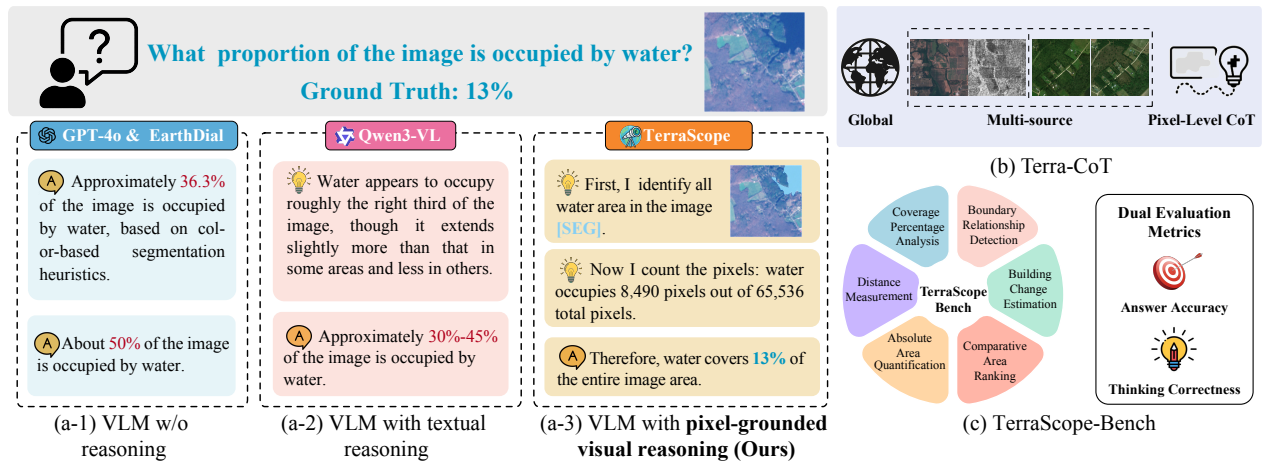


Figure 1. (a-1): The most common Vision Language Model (VLM) without reasoning directly outputs the wrong results. (a-2): Some solutions tried reasoning via textual Chain-of-Thought (CoT). (a-3): Our TerraScope, which takes the pixel-level grounding masks together with textual input, forming the interleaved CoT. (b): Our Terra-CoT 1M dataset. (c): Our TerraScope benchmark.

Abstract

Vision-language models (VLMs) have shown promise in earth observation (EO), yet they struggle with tasks that require grounding complex spatial reasoning in precise pixel-level visual representations. To address this problem, we introduce TerraScope, a unified VLM that delivers pixel-grounded geospatial reasoning with two key capabilities: (1) *modality-flexible reasoning*: it handles single-modality inputs (optical or SAR) and adaptively fuses different modalities into the reasoning process when both are available; (2) *multi-temporal reasoning*: it integrates temporal sequences for change analysis across multiple time points. In addition, we curate Terra-CoT, a large-scale dataset containing 1 million samples with pixel-level masks embedded in reasoning chains across multiple sources. We also propose TerraScope-Bench, the first benchmark for pixel-grounded geospatial reasoning with six sub-tasks that evaluates both answer accuracy and mask quality to ensure authentic pixel-grounded reasoning. Experiments show

that TerraScope significantly outperforms existing VLMs on pixel-grounded geospatial reasoning while providing interpretable visual evidence.

1. Introduction

Earth observation (EO) satellites continuously monitor our planet at unprecedented scales, generating vast imagery archives for environmental monitoring [49], disaster response [48], and resource management [16]. Traditional approaches [13, 19, 20, 45] to EO data analysis rely on task-specific models, limiting flexibility across diverse applications. Vision-language models (VLMs) offer a paradigm shift: unified models that understand both visual content and natural language, enabling flexible analysis through text-based interaction. Recent domain-adapted VLMs have demonstrated strong performance on standard EO tasks, including image captioning [30], visual question answering [29, 31, 43, 52], and visual grounding [15, 51, 54, 58], leveraging large-scale instruction tuning on remote sensing data.

However, state-of-the-art VLMs struggle with fine-grained geospatial reasoning requiring pixel-accurate spa-

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tial analysis. As illustrated in Fig. 1, leading general-purpose models (GPT-4o [34]), reasoning-capable models (Qwen3-VL [1]), and EO-specific variants (EarthDial [43]) all fail to provide accurate answers on tasks such as calculating coverage of a land-cover class given in an image. Recent multi-modal reasoning models [7, 56, 57] have shown promise by grounding visual regions before reasoning. However, they cannot directly transfer to EO due to two fundamental differences: (i) Unlike natural images with discrete objects, EO imagery depicts continuous spatial distributions where land cover types transition gradually. This continuous nature introduces substantial noise when coarse-grained grounding is used, hindering reasoning accuracy. (ii) EO analysis often involves multi-sensor and temporally evolving data. Optical imagery captures surface reflectance, SAR provides all-weather observation, and multi-temporal sequences reveal dynamic changes. However, existing VLMs struggle to effectively integrate such modality-flexible, time-varying data for EO reasoning within a single, unified framework.

To address these challenges, we present TerraScope, a comprehensive framework for pixel-grounded visual reasoning in EO. Building upon the recent paradigm of “*thinking with images*” [44], TerraScope embodies the principle of “*thinking with pixels*”: it explicitly localizes task-relevant regions and grounds each reasoning step in pixel-level visual evidence, rather than operating solely within the language domain. Prior VLMs for EO rely on external tools [4, 8, 25, 37] for reasoning. The incorporation of external tools substantially increases the model’s complexity and reduces controllability, making it difficult to achieve pixel-level, intrinsic reasoning. In contrast, TerraScope employs mixed decoders that jointly generate segmentation masks and reasoning traces. The language model autonomously decides when to trigger mask generation and interleave the resulting visual tokens into the reasoning process, enabling dynamic visual grounding throughout multi-step reasoning. Beyond single-date single-modality data, TerraScope supports two independent reasoning capabilities. First, for *multi-temporal reasoning*, it analyzes observations from multiple time points to deduce temporal changes based on evolving spatial patterns. Second, for *multi-modal reasoning*, when both optical and SAR data are available, it adaptively selects the most informative modality for each reasoning step through text-guided cross-attention, leveraging optical for spectral information in clear regions while relying on SAR for cloud-covered areas. To enable pixel-grounded reasoning at scale, we curate Terra-CoT, a 1M instruction-tuning dataset with pixel-level masks embedded in reasoning traces generated via an automated pipeline, covering global scenes across multi-source EO data. Additionally, existing EO benchmarks [23, 29, 46] primarily focus on visual perception

tasks and lack evaluation of fine-grained visual reasoning capabilities. We introduce TerraScope-Bench, a benchmark specifically designed for pixel-grounded geospatial reasoning. It comprises 3,837 expert-verified questions supporting flexible evaluation with optical-only, SAR-only, or joint optical-SAR data, across both single-date and multi-temporal scenarios. Beyond traditional VQA accuracy metrics, TerraScope-Bench introduces dual evaluation metrics that assess both answer correctness and segmentation mask quality, ensuring models genuinely ground reasoning in pixel-level visual evidence.

In summary, our contributions are threefold:

- We introduce **TerraScope**, a unified framework for pixel-grounded visual reasoning in EO. It grounds each reasoning step in precise segmentation masks for fine-grained, interpretable spatial analysis, supports multi-temporal change reasoning, and adaptively uses optical or SAR imagery.
- We curate **Terra-CoT**, a 1M instruction-tuning dataset with pixel-accurate masks embedded in reasoning traces, enabling scalable pixel-grounded training.
- We propose **TerraScope-Bench**, a benchmark of 3,837 expert-verified samples with dual metrics for answer accuracy and mask quality. Experiments on 11 models expose current limitations and demonstrate the effectiveness of TerraScope.

2. Related Works

Earth Observation VLMs. Recent advancements in general-purpose VLMs [18, 26, 59] have shown impressive capabilities across various tasks. However, their limited exposure to remote sensing imagery hinders performance on EO tasks. To address this gap, specialized EO-VLMs have emerged through domain-specific data curation and model adaptation. RSGPT [12] enriches captioning datasets to enhance LLaVA’s conversational abilities on satellite imagery. SkyEye-GPT [52] synthesizes 968K instruction samples for multi-task learning. Beyond image-level tasks, GeoChat [15], SkySenseGPT [31], and LHRS-Bot [33] incorporate visual grounding, region captioning, and reasoning. EarthGPT [54] introduces multi-sensor datasets spanning optical, SAR, and infrared modalities, while EarthMarker and EarthGPT-X [55] enable visual prompting interactions. GeoPixel [38] focuses on pixel-level grounding with grounded conversation datasets. EarthDial [43] scales multi-sensor data across multispectral, hyperspectral, and SAR to improve generalization. VHM [35] proposes datasets with both factual and deceptive questions to improve model honesty. Despite these advances, existing EO-VLMs still lack pixel-grounded reasoning capabilities required for fine-grained spatial analysis.

Earth Observation Benchmarks. The rapid development of EO-VLMs has stimulated dedicated evaluation bench-

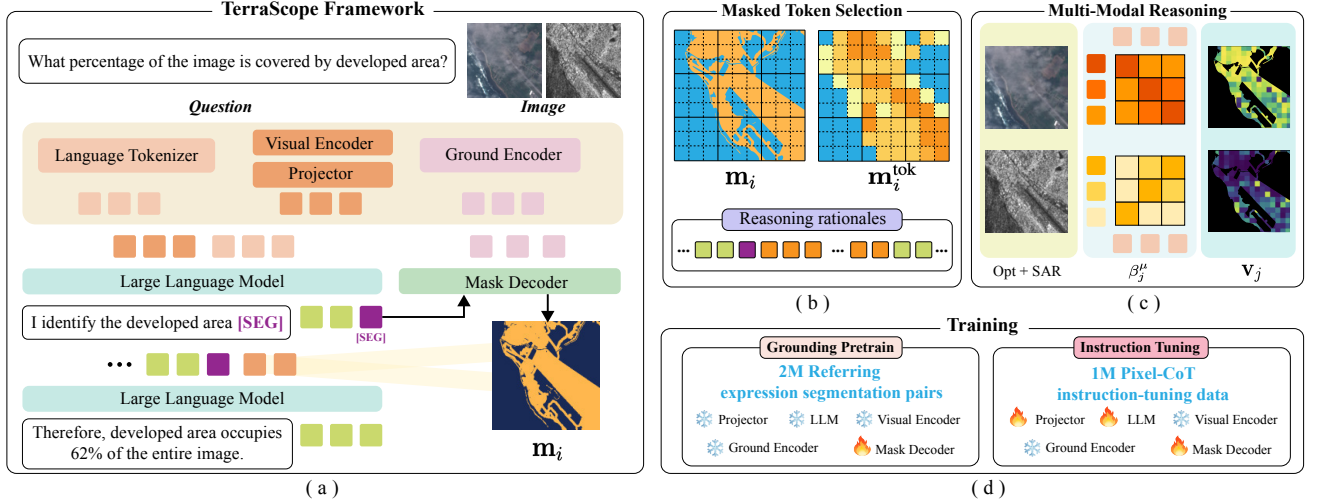


Figure 2. **Overview of TerraScope.** TerraScope generates textual reasoning tokens and segmentation masks in an interleaved manner, where masked visual features are injected at each reasoning step to ensure faithful pixel-grounded reasoning. TerraScope supports multi-modal and multi-temporal reasoning across EO data.

marks. RSVQA [29], LHRS-Bench [33], RSIEval [12], and VLEO-Bench [53] evaluate conversational capabilities including classification, captioning, and VQA. VRS-Bench [23] and GeoChat-Bench [15] incorporate region-level grounding for localization evaluation. XLRs-Bench [46] focuses on ultra-high-resolution imagery understanding. GeoBench-VLM [6] is a comprehensive benchmark covering multi-task and multi-sensor EO scenarios. DisasterM3 [48] proposes a bi-temporal benchmark spanning multiple hazards, sensors, and tasks. While recent benchmarks broaden the scope of sensors, tasks, and temporal settings, they still do not rigorously assess models’ capacity for pixel-accurate geospatial inference, leaving a gap in evaluating the precision needed for detailed spatial analysis.

Visual Chain-of-Thought. Recent works have explored grounding reasoning processes in visual content by interleaving visual evidence with textual reasoning chains. GRIT [7] interleaves the bounding box coordinates with the natural language reasoning for fine-grained counting. DeepEyes [57], Chain-of-Focus [56], and Mini-o3 [17] employ iterative zoom-in mechanisms that crop and analyze focused regions. VLM-R1 [39] and Visual-RFT [28] leverage reinforcement learning for visual grounding tasks. MintCoT [3] and ICoT [9] select relevant visual tokens through retrieval or attention mechanisms to compose multimodal rationales. However, these methods rely on coarse-grained spatial representations (bounding boxes, crops, or implicit token selection), which are inadequate for geospatial reasoning requiring pixel-level segmentation to capture continuous spatial distributions across multimodal data.

3. Method

In this section, we present the core components of TerraScope and outline how pixel-grounded visual reasoning is

formulated and implemented within our framework.

3.1. Overview

Geospatial reasoning demands fine-grained visual understanding that language-only reasoning cannot provide. In this context, we propose *pixel-grounded visual reasoning*, where models explicitly generate segmentation masks and ground reasoning in the selected masked visual space. Formally, let $f(\cdot)$ be a VLM composed of a text encoder f_T and a vision encoder f_V . Given a question Q and an image I , the text encoder produces $\mathbf{q} = f_T(Q)$ and the vision encoder produces $\mathbf{v} = f_V(I) \in \mathbb{R}^{N \times D}$, where N is the number of visual tokens and D is the feature dimension. Traditional VLMs then output an answer via language-only reasoning:

$$[\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_k, \mathbf{a}] = f(\mathbf{v}, \mathbf{q}), \quad (1)$$

where k is the number of reasoning steps, $\mathbf{r}_i$ denotes the i -th textual reasoning step, and $\mathbf{a}$ is the final answer. Pixel-grounded visual reasoning interleaves masked visual features with textual reasoning:

$$[\mathbf{r}_1, (\mathbf{m}_1, \mathbf{v}_1), \mathbf{r}_2, (\mathbf{m}_2, \mathbf{v}_2), \dots, \mathbf{r}_k, (\mathbf{m}_k, \mathbf{v}_k), \mathbf{a}] = f(\mathbf{v}, \mathbf{q}), \quad (2)$$

where at each reasoning step i , the model generates a segmentation mask $\mathbf{m}_i$ and selects masked visual features $\mathbf{v}_i$ from the identified regions. In the rest of this section, we first present the TerraScope architecture that enables joint generation of masks and reasoning (Sec. 3.2), then describe Terra-CoT, our instruction dataset with interleaved visual and textual traces (Sec. 3.3).

3.2. TerraScope Framework

As shown in Fig. 2, our TerraScope builds upon a vision-language architecture augmented with a pixel-level segmen-

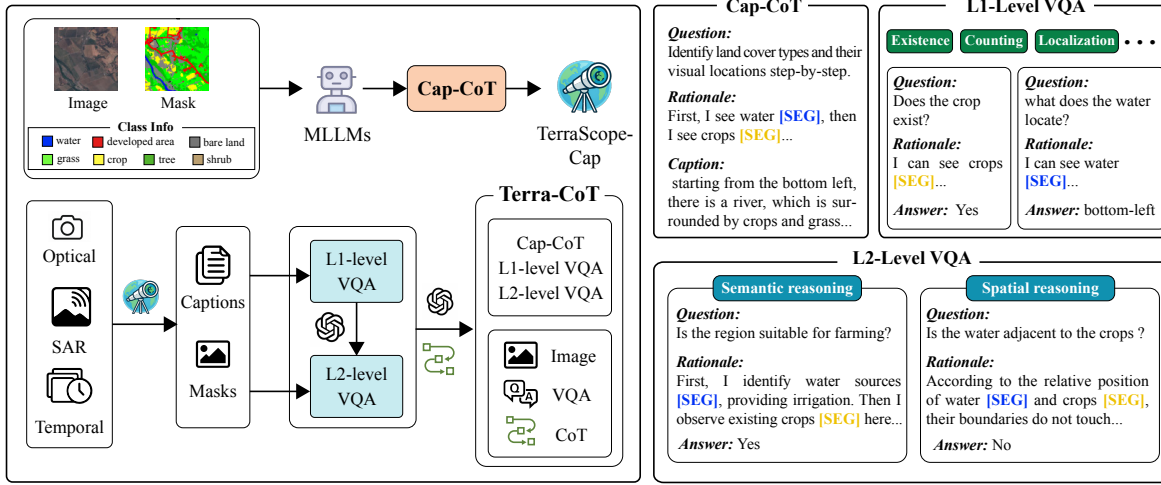


Figure 3. **Terra-CoT curation pipeline.** First, we generate Cap-CoT using ground truth masks and class labels to train an initial annotation model. Second, we use the trained model to annotate unlabeled data with pixel-accurate masks and captions. Third, based on the synthetic annotations, we apply hierarchical data synthesis to generate diverse reasoning questions with chain-of-thought traces at two levels: (L1) basic spatial grounding and (L2) complex multi-step reasoning including spatial and semantic tasks.

tation module, forming a unified framework that integrates visual grounding and language-based reasoning within a single model. Specifically, we leverage InternVL3 [59] as our base model, which dynamically splits single images into sub-tiles while processing multi-image inputs independently, thereby defining a unified pipeline to transform all data into a uniform format.

Pixel-Grounded Chain-of-Thought. The core innovation of TerraScope lies in the cooperative mechanism between dual decoders, which interleaves segmentation mask generation with text generation. Specifically, during the reasoning process, TerraScope monitors the language decoder’s autoregressive output and triggers the mask decoder upon detecting [SEG], which typically appears after mentions of key regions or objects. The mask decoder then predicts segmentation masks, from which masked visual tokens are selected and injected into the reasoning sequence to guide subsequent generation. For example, when answering “Which is larger, water or road?”, the model generates “I first identify water regions [SEG]...then road regions [SEG]” and derives the answer by comparing their masked visual features.

As shown in Fig. 2 (b), to inject high-quality visual representations corresponding to the generated mask into reasoning traces, we first align the mask $\mathbf{m}_i$ with the visual encoder’s dynamic patch layout by resizing it to the token grid resolution $(n \cdot s) \times (m \cdot s)$, where the image is split into $n \times m$ patches with each patch producing $s \times s$ tokens ($s = 16$ for InternVL). To handle partial overlap between the pixel-level mask and token grid, we select a visual token if the mask covers more than 50% of its corresponding spatial region. For the masked region, we extract the selected visual features as:

$$\mathbf{v}_i = \{\mathbf{v}_j \mid \mathbf{m}_i^{\text{tok}}[j] = 1, j \in [1, N]\} \quad (3)$$

where $\mathbf{v}_j$ denotes the j -th visual token in the feature map, and $\mathbf{m}_i^{\text{tok}}$ is the token-level mask derived from $\mathbf{m}_i$ by resizing to the token grid. The selected visual features $\mathbf{v}_i$ are then projected and flattened into a 1D sequence aligned with text embeddings, and fed into the LLM to resume autoregressive text generation conditioned on the KV cache of previously generated tokens.

Multi-Modal and Temporal Reasoning. Unlike single-image understanding, EO data often involves multi sources including optical-SAR pairs and temporal sequences. TerraScope handles these diverse scenarios through its flexible pixel-grounded reasoning framework.

For optical-SAR pairs, the model must identify complementary features, leveraging optical imagery for spectral information under clear conditions while relying on SAR for cloud-covered regions. We achieve this through text-guided, token-level modality selection. As shown in Fig. 2 (c), given optical and SAR images processed independently through the vision encoder to obtain visual features $\mathbf{v}_{\text{opt}}$ and $\mathbf{v}_{\text{SAR}}$, and question embeddings $\mathbf{q}$ from the text tokenizer with length L , we compute cross-attention between text and each visual modality, then aggregate across text tokens to obtain text-relevance scores:

$$\beta_j^\mu = \frac{1}{L} \sum_{\ell=1}^L \text{Softmax} \left(\frac{\mathbf{v}_j^\mu \mathbf{q}^\top}{\sqrt{D}} \right)_{j\ell}, \quad \mu \in \{\text{opt}, \text{SAR}\} \quad (4)$$

where β_j^μ denotes the relevance score of the j -th visual token to the question for modality μ . When selecting masked visual features $\mathbf{v}_i$, we select features from the modality with a higher relevance score for each token position:

$$\mathbf{v}_j = \begin{cases} \mathbf{v}_j^{\text{opt}} & \text{if } \beta_j^{\text{opt}} > \beta_j^{\text{SAR}} \\ \mathbf{v}_j^{\text{SAR}} & \text{otherwise} \end{cases}, \quad \forall j \text{ where } \mathbf{m}_i^{\text{tok}}[j] = 1 \quad (5)$$

This dynamic, spatially adaptive mechanism leverages the

complementarity of paired EO data to boost reasoning.

For temporal sequences, a critical challenge is temporal disambiguation: when reasoning involves multiple observations, each [SEG] token must specify (1) which temporal image the mask decoder should segment from, and (2) from which image to extract the masked visual tokens. To address this, we incorporate explicit temporal indicators in the format “Image: t_i ” before each [SEG] token. When the language decoder generates these signals, the mask decoder segments from image t_i and the feature extraction module samples visual tokens from $\mathbf{v}^{(t_i)}$. The model learns to generate timestamps from our Terra-CoT dataset, which contains temporally grounded reasoning traces paired with frame-specific masks (Sec. 3.3).

Training. We train TerraScope in two stages using supervised fine-tuning. We first train on 2M referring expression segmentation pairs to establish basic grounding capability. We then fine-tune on 1M Terra-CoT samples to incentivize pixel-grounded visual reasoning ability. During training, we extract masked visual features from ground truth masks and interleave them into the sequence at positions following [SEG] tokens. The training objective combines language modeling loss (cross-entropy on text and [SEG] tokens, excluding injected visual features) and segmentation loss (Dice loss and pixel-wise cross-entropy):

$$\mathcal{L} = \mathcal{L}_{\text{LM}} + \lambda \mathcal{L}_{\text{seg}}, \quad (6)$$

where we set $\lambda = 0.5$ to balance both objectives.

3.3. Terra-CoT Dataset

Curating pixel-grounded visual CoT data is non-trivial: existing EO datasets provide either segmentation labels [5, 50] or VQA pairs [29], but not both with reasoning traces. We address this with a two-stage automated pipeline enabling large-scale pixel-grounded reasoning data.

Grounded Captioning with Chain-of-Thought. We leverage existing datasets with semantic annotations [5, 22, 50] to construct pixel-grounded captioning data with reasoning traces (Cap-CoT). As shown in Fig. 3, we prompt a large multimodal model with an image where distinct land-cover categories are highlighted using colored masks and labeled accordingly. The model is instructed to produce detailed captions that explicitly reference these masked regions throughout its reasoning. This process yields 250K Cap-CoT samples, used to both train TerraScope and build an intermediate annotator, TerraScope-Cap, capable of generating pixel-grounded captions for unlabeled imagery.

Hierarchical Data Synthesis. Using TerraScope-Cap trained on Cap-CoT, we annotate images from diverse sources (optical, SAR, temporal) covering global regions with multi-category pixel-level labels (statistics in Appendix). Based on these annotations, we synthesize Terra-CoT through a two-level hierarchical process.

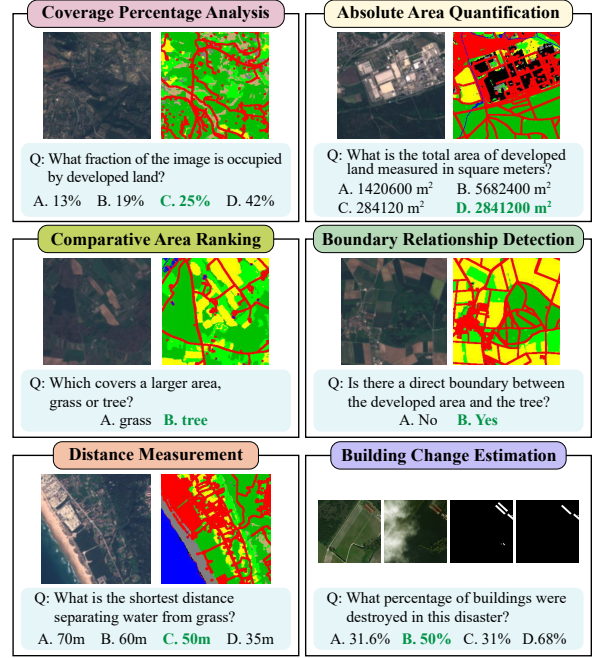


Figure 4. Examples of TerraScope-Bench.

Level 1 (L1): Basic spatial grounding. We generate template-based questions for randomly selected categories, covering fundamental spatial tasks such as existence verification, object counting, localization, area quantification, and boundary detection. For each question, we synthesize pixel-grounded reasoning traces using segmentation labels to explain the spatial analysis process.

Level 2 (L2): Complex multi-step reasoning. We prompt an LLM to compose multiple L1 questions into complex reasoning tasks of two types: (1) *L2-Spatial* requires cross-entity spatial analysis such as relationship inference (e.g., “Is the water adjacent to the crops?”); (2) *L2-Semantic* requires domain knowledge beyond visual observation such as land suitability assessment (e.g., “Is the region suitable for farming?”). For both types, the LLM synthesizes reasoning traces combining visual evidence with spatial or semantic analysis. This hierarchical process produces 1M Terra-CoT samples with diverse reasoning abilities.

4. TerraScope-Bench

EO imagery above 10m resolution presents unique challenges: individual objects span only a few pixels, and land-use boundaries become ambiguous, making precise pixel-level spatial reasoning essential. However, existing benchmarks (e.g., BigEarthNet [5], ChatEarthNet [50],) emphasize coarse-grained tasks such as scene classification and image captioning that depend primarily on global visual cues. As a result, they fail to adequately assess VLMs’ fine-grained reasoning capabilities, allowing models to perform well without genuine spatial understanding.

To address these limitations, we introduce **TerraScope-Bench**, a benchmark comprising 3,837 carefully curated

Model	Size	TerraScope-Bench							Landsat30AU				DisasterM3		
		CA	AQ	CR	BRD	DM	BCE	Avg.	APR	NUM	SRI	Avg.	BDC	DRE	Avg.
General VLMs															
GPT-4o [†] [34]	-	27.6	25.4	54.3	75.3	22.5	27.1	38.7	-	-	-	-	24.2	21.4	22.8
LLaVA-OV [18]	7B	28.0	21.2	56.6	75.9	19.4	23.7	37.5	39.4	46.6	85.1	57.0	26.4	24.2	25.3
Qwen2.5-VL [1]	7B	25.3	33.5	55.7	67.7	23.3	25.7	38.5	29.8	53.1	92.8	58.6	34.2	29.3	31.8
InternVL3 [59]	8B	22.3	26.3	57.2	67.0	18.6	24.3	36.0	31.4	42.4	90.6	54.8	30.3	24.1	27.2
GLM-4.1V-Think [‡] [11]	9B	24.8	57.1	55.2	58.4	23.3	29.5	41.4	45.7	58.6	70.0	58.1	-	-	-
Qwen3-VL-Think [‡] [1]	8B	29.0	47.8	57.9	67.8	25.6	31.9	43.3	42.8	60.2	92.0	65.0	36.8	28.2	32.5
EO-Specific VLMs															
GeoChat [15]	7B	24.8	19.5	49.6	69.2	5.4	-	33.7	31.1	41.8	86.2	53.0	-	-	-
TeoChat [14]	7B	25.6	17.8	55.8	55.8	8.5	22.6	31.0	30.2	59.6	87.1	59.0	22.5	23.3	22.9
LHRS-bot [33]	7B	13.7	24.3	54.0	28.4	12.4	-	26.6	63.5	12.5	82.6	52.9	-	-	-
EarthDial [43]	4B	26.3	24.1	54.4	69.2	20.2	23.6	36.3	23.5	43.6	51.2	39.4	30.2	20.8	25.5
EarthMind [42]	4B	26.1	42.2	52.2	73.3	38.1	20.8	42.1	-	-	-	-	-	-	-
Fine-tuned VLMs															
InternVL3 [59]	8B	67.1	63.2	60.0	67.8	40.0	31.0	54.9	55.3	56.6	90.8	67.6	42.2	30.1	36.1
GLM-4.1V-Think [‡] [11]	9B	67.8	68.1	65.5	70.2	51.1	34.7	59.6	63.4	60.5	80.0	68.0	45.6	32.0	38.8
TerraScope	8B	73.2	70.2	71.8	80.0	65.9	52.1	68.9	69.8	60.8	91.1	73.9	54.1	38.9	46.5

Table 1. Quantitative performance of TerraScope on TerraScope-Bench (optical only), Landsat30AU and DisasterM3. “Avg” is the average performance of multiple-choice tasks. Bold means the best performance. [†] denotes proprietary models. [‡] denotes reasoning models. Fine-tuned VLMs are fine-tuned on our proposed Terra-CoT dataset.

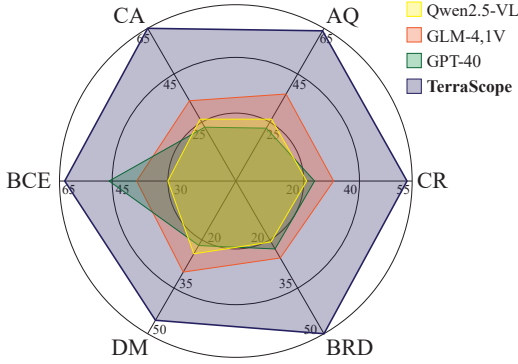


Figure 5. Grounding IoU performance of different models.

samples from the test sets of existing datasets [5, 10, 50]. As shown in Fig. 4, our benchmark encompasses six task categories: Coverage Percentage Analysis (855), Absolute Area Quantification (855), Distance Measurement (129), Comparative Area Ranking (855), Boundary Relationship Detection (855), and Building Change Estimation (288).

We leverage pixel-level segmentation annotations to automatically generate question-answer pairs. For each sample, we compute spatial properties from segmentation masks, including coverage ratios, absolute areas, inter-object distances, and boundary relationships, to derive ground-truth answers. Questions are generated via templates to ensure diverse phrasing, then rephrased by an LLM to create natural variations and plausible distractors for multiple-choice format. Finally, human experts review the dataset to filter samples with erroneous masks. Unlike existing benchmarks that only assess final answer accuracy, TerraScope-Bench evaluates both response correctness and spatial reasoning quality using IoU-based segmentation metrics, verifying whether models attend to the correct regions during their reasoning process.

5. Experiments

Implementation details. Following the two-stage training strategy in Sec. 3.2, we first perform grounding pre-training where the vision encoder, projector, and LLM are frozen, and only the mask decoder is trained ($\text{lr}=2\text{e-}5$, batch size=8). In the second stage, we unfreeze the projector and mask decoder for full training and fine-tune the LLM via LoRA ($\text{lr}=1\text{e-}5$, batch size=2). The vision encoder is kept frozen during training. All experiments run on NVIDIA H200-141GB GPUs, with additional dataset and hyperparameter details provided in the Appendix.

Benchmarks. Beyond our proposed TerraScope-Bench, we evaluate TerraScope on two representative EO benchmarks in zero-shot settings to demonstrate its generalization capability. **LandSat30-AU** [32] features 30-meter resolution imagery with challenging reasoning subtasks; we report results on four tasks requiring fine-grained geospatial reasoning: Agro-Phenology Reasoning (APR), Numerosity Estimation (NUM), and Spatial-Relationship Inference (SRI). **DisasterM3** [48] is a bi-temporal disaster assessment benchmark with pre- and post-event image pairs covering multi-hazard scenarios across multiple sensors; we evaluate on Damaged Building Counting (DBC) and Damaged Road Area Estimation (DRE).

5.1. Main Results

We present the performance of TerraScope on several EO benchmarks in Tab. 1, where we evaluate 11 VLMs on TerraScope-Bench, including proprietary models and both general and EO-specific models. Additionally, we fine-tune InternVL3 and GLM-4.1V-Think on our Terra-CoT dataset to show its effectiveness. We highlight several key findings: **(1) Pixel-grounded reasoning remains challenging.** Existing VLMs struggle with fine-grained geospatial reason-

Model	TerraBen.	Landsat.	Disaster.
Original	33.8	45.7	23.6
Textual CoT w/o Seg.	58.7	56.5	32.9
Textual CoT with Seg.	60.6	58.9	35.8
Random-Mask CoT	43.2	53.8	32.6
Box CoT	62.8	70.5	43.9
TerraScope	68.9	73.9	46.5

Table 2. Ablation study on the effect of different CoT strategies for pixel-grounded visual reasoning. “Original” denotes the base TerraScope model after pretraining, upon which we fine-tune with different CoT variations via SFT.

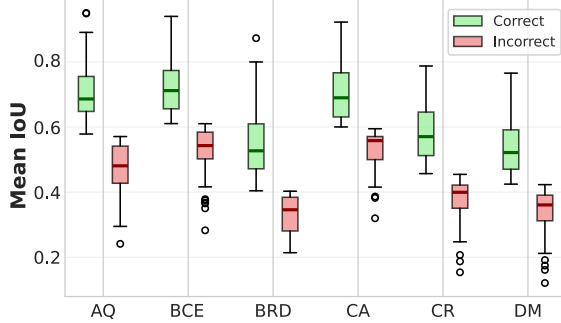


Figure 6. IoU distribution for correct vs. incorrect predictions.

ing, particularly on tasks requiring precise spatial analysis such as area percentage estimation. Both proprietary and open-source models achieve near-random performance, indicating the necessity of pixel-level grounding.

(2) **EO-specific models show limited advantages.** Despite training on large-scale EO data, EO-specific VLMs do not significantly outperform general VLMs on TerraScope-Bench. We hypothesize that this is because existing EO datasets predominantly feature high-resolution imagery (<5m), limiting models’ ability to handle lower-resolution data prevalent in real-world applications.

(3) **Reasoning models perform better but lack visual grounding.** Models with explicit reasoning capabilities show stronger performance, especially on tasks requiring external knowledge like Absolute Area Quantification. However, their reasoning remains purely textual without grounding in pixel-level visual evidence, leading to hallucinations and insufficient fine-grained spatial perception.

(4) **Terra-CoT effectively improves model performance.** Fine-tuning general VLMs (e.g., InternVL3, GLM-4.1V-Think) on our Terra-CoT dataset leads to substantial performance gains across all tasks, demonstrating the effectiveness of our pixel-grounded reasoning data. However, challenging tasks like distance measurement (DM) and building change estimation (BCE) remain difficult, suggesting that data alone is insufficient and specialized architectural designs for pixel-grounded reasoning are necessary.

(5) **TerraScope achieves strong performance and generalization.** Our framework, which grounds reasoning in fine-grained visual perception, achieves the best results on TerraScope-Bench while demonstrating strong generaliza-

Model	CA	AQ	CR	BRD	DM
No Fusion	73.2	70.2	71.8	80.0	65.9
Concat.	74.5	71.6	73.0	81.2	67.4
Text-guided (test only.)	72.3	69.0	66.7	78.8	63.6
Text-guided (train + test)	74.3	70.9	72.7	80.7	68.2

Table 3. Ablation study of multi-modal reasoning.

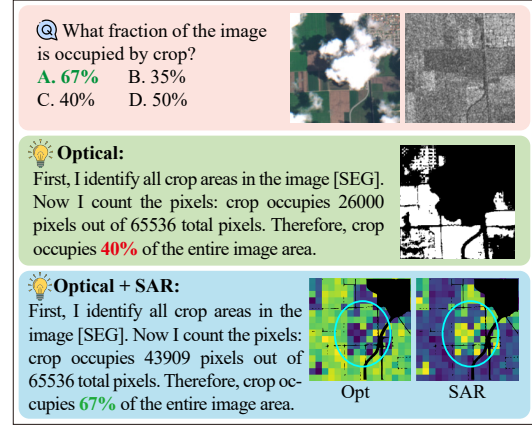


Figure 7. Visualizations of multi-modal reasoning.

tion to LandSat30-AU and DisasterM3.

(6) **TerraScope provides interpretable reasoning.** Beyond answer accuracy, TerraScope-Bench evaluates the reasoning process by measuring segmentation IoU against ground truth. As shown in Fig. 5, TerraScope not only produces correct answers but also generates faithful reasoning traces with accurate spatial grounding, outperforming other grounding-capable models.

5.2. Ablation Studies

We conduct extensive ablation studies to analyze TerraScope’s effectiveness regarding its pixel-grounded visual reasoning mechanism and multi-modal reasoning, in which more details can be seen in Appendix.

Effectiveness of pixel-grounded visual reasoning. To verify that pixel-level grounding benefits reasoning, we compare several variants in Tab. 2. First, we train models with textual chain-of-thought only, where visual tokens are not interleaved into reasoning steps: either freezing the mask decoder (*Textual CoT w/o Seg.*) or training it with ground truth masks as auxiliary supervision (*Textual CoT w/ Seg.*). Results show that auxiliary segmentation training implicitly improves reasoning even when visual tokens are absent from the reasoning sequence, demonstrating the benefit of our joint training design.

Second, we examine the importance of mask-guided token selection. *Random-Mask CoT*, which interleaves randomly selected visual tokens at each reasoning step without mask prediction, performs worse than textual CoT, likely due to irrelevant visual information hindering reasoning. *Box CoT* uses the minimal bounding rectangles of predicted masks to select visual tokens, rather than using pre-

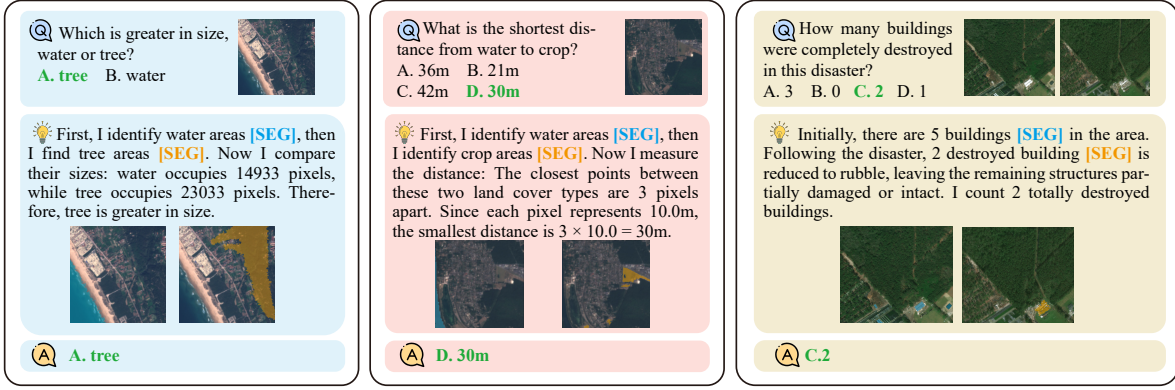


Figure 8. Visualization of TerraScope.

cise segmentation masks. This coarser grounding also underperforms TerraScope, especially on TerraScope-Bench and LandSat30-AU where land cover regions have irregular boundaries and shapes. These results confirm that precise pixel-level grounding through segmentation masks is essential for effective visual reasoning in EO.

Beyond final answer accuracy, we analyze the relationship between intermediate segmentation quality (measured by mean IoU with ground truth masks) and answer correctness. As shown in Fig. 6, samples with higher segmentation quality are significantly more likely to produce correct answers. Specifically, correct predictions achieve mean IoU of 0.628, substantially higher than incorrect predictions (0.443). The strong Pearson correlation ($r = 0.607$, $p < 0.001$) holds consistently across all task types ($r = 0.70$ - 0.80), demonstrating that accurate pixel-level visual grounding is essential for correct geospatial reasoning.

Effectiveness of multi-modal reasoning. Beyond optical imagery, TerraScope can reason across optical and SAR modalities through text-guided modality selection. To validate its effectiveness, we compare several settings in Tab. 3: (1) *No fusion*: using only optical images; (2) *Concat*: concatenating optical and SAR features and interleaving the concatenated features into each reasoning step; (3) *Text-guided (test only)*: enabling modality selection only during inference; (4) *Text-guided (train+test)*: enabling modality selection during both training and inference.

Results show that any form of multi-modal fusion substantially improves over the optical-only baseline. While *Concat* achieves slightly higher accuracy than text-guided selection, our approach offers a critical advantage: it reduces context length by selecting only the relevant modality rather than processing both, improving efficiency while maintaining competitive performance. Importantly, training with modality selection is essential—enabling it only at test time yields no improvement, demonstrating that the model must learn when and how to leverage different modalities.

Fig. 7 illustrates the effectiveness of our multi-modal reasoning through a cloud-contaminated case. The figure reveals two key advantages: (1) **Improved segmentation through multi-modal fusion**: When using optical imagery

alone, cloud cover causes severe segmentation errors in the masked regions. By fusing SAR data, which penetrates clouds, TerraScope produces accurate segmentation masks. (2) **Adaptive modality selection for reasoning**: Our text-guided selection mechanism adaptively chooses between optical and SAR based on data quality: it prioritizes optical tokens for cloud-free regions with reliable spectral information while selecting SAR tokens for cloud-covered areas where optical data is corrupted. This dual advantage enables accurate reasoning under challenging conditions.

6. Qualitative Results

Fig. 8 presents representative examples that demonstrating TerraScope’s pixel-grounded reasoning capabilities across three challenging tasks: area percentage estimation, distance measurement, and temporal counting VQA. These cases illustrate TerraScope’s dual strengths: (1) **Structured reasoning**: decomposing complex spatial questions into interpretable sub-steps through textual chain-of-thought, and (2) **Accurate visual grounding**: generating precise segmentation masks for relevant regions at each reasoning step. By grounding numerical computations in pixel-accurate visual evidence, TerraScope produces interpretable answers with transparent reasoning traces. Additional qualitative results and failure case analysis are provided in the Appendix.

7. Conclusion

In this paper, we presented *TerraScope*, a unified vision-language framework for pixel-grounded geospatial reasoning in earth observation. By generating segmentation masks alongside reasoning traces, TerraScope achieves precise and interpretable spatial analysis, supporting multi-temporal change analysis and adaptive reasoning across optical and SAR modalities. We curated *Terra-CoT*, a 1M instruction-tuning dataset with pixel-accurate masks embedded in reasoning chains, and introduced *TerraScope-Bench*, the first benchmark for pixel-grounded geospatial reasoning. Extensive experiments validate the effectiveness of our approach across diverse geospatial reasoning tasks.

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References

- [1] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, et al. Qwen2. 5-vl technical report. *arXiv preprint arXiv:2502.13923*, 2025. 2, 6
- [2] Zechen Bai, Pichao Wang, Tianjun Xiao, Tong He, Zongbo Han, Zheng Zhang, and Mike Zheng Shou. Hallucination of multimodal large language models: A survey. *arXiv preprint arXiv:2404.18930*, 2024. 1
- [3] Xinyan Chen, Renrui Zhang, Dongzhi Jiang, Aojun Zhou, Shilin Yan, Weifeng Lin, and Hongsheng Li. Mint-cot: Enabling interleaved visual tokens in mathematical chain-of-thought reasoning. *arXiv preprint arXiv:2506.05331*, 2025. 3, 1
- [4] Yuxing Chen, Weijie Wang, Sylvain Lobry, and Camille Kurtz. An llm agent for automatic geospatial data analysis. *arXiv preprint arXiv:2410.18792*, 2024. 2
- [5] Kai Norman Clasen, Leonard Hackel, Tom Burgert, Gencer Sumbul, Begüm Demir, and Volker Markl. reBEN: Refined bigearthnet dataset for remote sensing image analysis. In *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2025. 5, 6, 10
- [6] Muhammad Sohail Danish, Muhammad Akhtar Munir, Syed Roshaan Ali Shah, Kartik Kuckreja, Fahad Shahbaz Khan, Paolo Fraccaro, Alexandre Lacoste, and Salman Khan. Geobench-vlm: Benchmarking vision-language models for geospatial tasks. *arXiv preprint arXiv:2411.19325*, 2024. 3
- [7] Yue Fan, Xuehai He, Diji Yang, Kaizhi Zheng, Ching-Chen Kuo, Yuting Zheng, Sravana Jyothi Narayanaraju, Xinze Guan, and Xin Eric Wang. Grit: Teaching mllms to think with images. *arXiv preprint arXiv:2505.15879*, 2025. 2, 3, 1
- [8] Peilin Feng, Zhutao Lv, Junyan Ye, Xiaolei Wang, Xinjie Huo, Jinhua Yu, Wanghan Xu, Wenlong Zhang, Lei Bai, Conghui He, et al. Earth-agent: Unlocking the full landscape of earth observation with agents. *arXiv preprint arXiv:2509.23141*, 2025. 2
- [9] Jun Gao, Yongqi Li, Ziqiang Cao, and Wenjie Li. Interleaved-modal chain-of-thought. In *CVPR*, pages 19520–19529, 2025. 3, 1
- [10] Ritwik Gupta, Richard Hosfelt, Sandra Sajeev, Nirav Patel, Bryce Goodman, Jigar Doshi, Eric Heim, Howie Choset, and Matthew Gaston. xbd: A dataset for assessing building damage from satellite imagery. *arXiv preprint arXiv:1911.09296*, 2019. 6
- [11] Wenyi Hong, Wenmeng Yu, Xiaotao Gu, Guo Wang, Guobing Gan, Haomiao Tang, Jiale Cheng, Ji Qi, Junhui Ji, Li-hang Pan, et al. Glm-4.1 v-thinking: Towards versatile multimodal reasoning with scalable reinforcement learning. *arXiv e-prints*, pages arXiv-2507, 2025. 6, 9
- [12] Yuan Hu, Jianlong Yuan, Congcong Wen, Xiaonan Lu, Yu Liu, and Xiang Li. Rsgpt: A remote sensing vision language model and benchmark. *ISPRS Journal of Photogrammetry and Remote Sensing*, 224:272–286, 2025. 2, 3
- [13] Shiqi Huang, Shuting He, Huaiyuan Qin, and Bihan Wen. Score: Scene context matters in open-vocabulary remote sensing instance segmentation. In *ICCV*, pages 12559–12569, 2025. 1
- [14] Jeremy Andrew Irvin, Emily Ruoyu Liu, Joyce Chuyi Chen, Ines Dormoy, Jinyoung Kim, Samar Khanna, Zhuo Zheng, and Stefano Ermon. TeoChat: A large vision-language assistant for temporal earth observation data. *arXiv preprint arXiv:2410.06234*, 2024. 6
- [15] Kartik Kuckreja, Muhammad Sohail Danish, Muzammal Naseer, Abhijit Das, Salman Khan, and Fahad Shahbaz Khan. GeoChat: Grounded large vision-language model for remote sensing. In *CVPR*, pages 27831–27840, 2024. 1, 2, 3, 6
- [16] Sandeep Kumar, Ram Swaroop Meena, Seema Sheoran, Chetan Kumar Jangir, Manoj Kumar Jhariya, Arnab Banerjee, and Abhishek Raj. Remote sensing for agriculture and resource management. In *Natural resources conservation and advances for sustainability*, pages 91–135. Elsevier, 2022. 1
- [17] Xin Lai, Junyi Li, Wei Li, Tao Liu, Tianjian Li, and Hengshuang Zhao. Mini-o3: Scaling up reasoning patterns and interaction turns for visual search. *arXiv preprint arXiv:2509.07969*, 2025. 3
- [18] Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Peiyuan Zhang, Yanwei Li, Ziwei Liu, et al. Llava-onevision: Easy visual task transfer. *arXiv preprint arXiv:2408.03326*, 2024. 2, 6
- [19] Erzhu Li, Junshi Xia, Peijun Du, Cong Lin, and Alim Samat. Integrating multilayer features of convolutional neural networks for remote sensing scene classification. *TGRS*, 55(10): 5653–5665, 2017. 1
- [20] Kaiyu Li, Ruixun Liu, Xiangyong Cao, Xueru Bai, Feng Zhou, Deyu Meng, and Zhi Wang. Segearth-ov: Towards training-free open-vocabulary segmentation for remote sensing images. In *CVPR*, pages 10545–10556, 2025. 1
- [21] Qi Li and Xinchao Wang. Sponge tool attack: Stealthy denial-of-efficiency against tool-augmented agentic reasoning. *arXiv preprint arXiv:2601.17566*, 2026. 1
- [22] Xue Li, Guo Zhang, Hao Cui, Shasha Hou, Shun Yao Wang, Xin Li, Yujia Chen, Zhijiang Li, and Li Zhang. Mcanet: A joint semantic segmentation framework of optical and sar images for land use classification. *International Journal of Applied Earth Observation and Geoinformation*, 106: 102638, 2022. 5
- [23] Xiang Li, Jian Ding, and Mohamed Elhoseiny. Vrsbench: A versatile vision-language benchmark dataset for remote sensing image understanding. *arXiv preprint arXiv:2406.12384*, 2024. 2, 3
- [24] Xixun Lin, Yucheng Ning, Jingwen Zhang, Yan Dong, Yilong Liu, Yongxuan Wu, Xiaohua Qi, Nan Sun, Yanmin

- Shang, Kun Wang, et al. Llm-based agents suffer from hallucinations: A survey of taxonomy, methods, and directions. *arXiv preprint arXiv:2509.18970*, 2025. 1
- [25] Chenyang Liu, Keyan Chen, Haotian Zhang, Zipeng Qi, Zhengxia Zou, and Zhenwei Shi. Change-agent: Towards interactive comprehensive remote sensing change interpretation and analysis. *TGRS*, 2024. 2
- [26] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. *NeurIPS*, 36:34892–34916, 2023. 2
- [27] Hao Liu, Yongjie Zheng, Yuhan Kang, Mingyang Zhang, Maoguo Gong, and Lorenzo Bruzzone. Balanced diffusion-guided fusion for multimodal remote sensing classification. *arXiv preprint arXiv:2509.23310*, 2025. 1
- [28] Ziyu Liu, Zeyi Sun, Yuhang Zang, Xiaoyi Dong, Yuhang Cao, Haodong Duan, Dahua Lin, and Jiaqi Wang. Visual-rft: Visual reinforcement fine-tuning. *arXiv preprint arXiv:2503.01785*, 2025. 3
- [29] Sylvain Lobry, Diego Marcos, Jesse Murray, and Devis Tuia. Rsvqa: Visual question answering for remote sensing data. *TGRS*, 58(12):8555–8566, 2020. 1, 2, 3, 5, 10
- [30] Xiaoqiang Lu, Binqiang Wang, Xiangtao Zheng, and Xuelong Li. Exploring models and data for remote sensing image caption generation. *TGRS*, 56(4):2183–2195, 2017. 1
- [31] Junwei Luo, Zhen Pang, Yongjun Zhang, Tingzhu Wang, Linlin Wang, Bo Dang, Jiangwei Lao, Jian Wang, Jingdong Chen, Yihua Tan, et al. Skysensegpt: A fine-grained instruction tuning dataset and model for remote sensing vision-language understanding. *arXiv preprint arXiv:2406.10100*, 2024. 1, 2
- [32] Sai Ma, Zhuang Li, and John A Taylor. Landsat30-au: A vision-language dataset for australian landsat imagery. *arXiv preprint arXiv:2508.03127*, 2025. 6
- [33] Dilxat Muhtar, Zhenshi Li, Feng Gu, Xueliang Zhang, and Pengfeng Xiao. Lhrs-bot: Empowering remote sensing with vgi-enhanced large multimodal language model. In *ECCV*, pages 440–457. Springer, 2024. 2, 3, 6
- [34] OpenAI. Gpt-4o, 2024. 2, 6
- [35] Chao Pang, Xingxing Weng, Jiang Wu, Jiayu Li, Yi Liu, Jiaxing Sun, Weijia Li, Shuai Wang, Litong Feng, Gui-Song Xia, et al. Vhm: Versatile and honest vision language model for remote sensing image analysis. In *AAAI*, pages 6381–6388, 2025. 2
- [36] Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, et al. Sam 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024. 2
- [37] Akashah Shabbir, Muhammad Akhtar Munir, Akshay Dudhane, Muhammad Umer Sheikh, Muhammad Haris Khan, Paolo Fraccaro, Juan Bernabe Moreno, Fahad Shahbaz Khan, and Salman Khan. Thinkgeo: Evaluating tool-augmented agents for remote sensing tasks. *arXiv preprint arXiv:2505.23752*, 2025. 2
- [38] Akashah Shabbir, Mohammed Zumri, Mohammed Benmamoun, Fahad S Khan, and Salman Khan. Geopixel: Pixel grounding large multimodal model in remote sensing. *ICML*, 2025. 2
- [39] Haozhan Shen, Peng Liu, Jingcheng Li, Chunxin Fang, Yibo Ma, Jiajia Liao, Qiaoli Shen, Zilun Zhang, Kangjia Zhao, Qianqian Zhang, et al. Vlm-r1: A stable and generalizable r1-style large vision-language model. *arXiv preprint arXiv:2504.07615*, 2025. 3
- [40] Yan Shu, Hangui Lin, Yexin Liu, Yan Zhang, Gangyan Zeng, Yan Li, Yu Zhou, Ser-Nam Lim, Harry Yang, and Nicu Sebe. When semantics mislead vision: Mitigating large multimodal models hallucinations in scene text spotting and understanding. *arXiv preprint arXiv:2506.05551*, 2025. 1
- [41] Yan Shu, Zheng Liu, Peitian Zhang, Minghao Qin, Junjie Zhou, Zhengyang Liang, Tiejun Huang, and Bo Zhao. Video-xl: Extra-long vision language model for hour-scale video understanding. In *CVPR*, pages 26160–26169, 2025. 1
- [42] Yan Shu, Bin Ren, Zhitong Xiong, Danda Pani Paudel, Luc Van Gool, Begüm Demir, Nicu Sebe, and Paolo Rota. Earthmind: Leveraging cross-sensor data for advanced earth observation interpretation with a unified multimodal llm. *arXiv preprint arXiv:2506.01667*, 2025. 6
- [43] Sagar Soni, Akshay Dudhane, Hiyam Debary, Mustansar Fiaz, Muhammad Akhtar Munir, Muhammad Sohail Danish, Paolo Fraccaro, Campbell D Watson, Levente J Klein, Fahad Shahbaz Khan, et al. Earthdial: Turning multi-sensory earth observations to interactive dialogues. *arXiv preprint arXiv:2412.15190*, 2024. 1, 2, 6
- [44] Zhaochen Su, Peng Xia, Hangyu Guo, Zhenhua Liu, Yan Ma, Xiaoye Qu, Jiaqi Liu, Yanshu Li, Kaide Zeng, Zhengyuan Yang, et al. Thinking with images for multimodal reasoning: Foundations, methods, and future frontiers. *arXiv preprint arXiv:2506.23918*, 2025. 2
- [45] Hao Sun, Siyuan Li, Xiangtao Zheng, and Xiaoqiang Lu. Remote sensing scene classification by gated bidirectional network. *TGRS*, 58(1):82–96, 2019. 1
- [46] Fengxiang Wang, Hongzhen Wang, Mingshuo Chen, Di Wang, Yulin Wang, Zonghao Guo, Qiang Ma, Long Lan, Wenjing Yang, Jing Zhang, et al. Xlrs-bench: Could your multimodal llms understand extremely large ultra-high-resolution remote sensing imagery? *arXiv preprint arXiv:2503.23771*, 2025. 2, 3
- [47] Jiacong Wang, Zijian Kang, Haochen Wang, Haiyong Jiang, Jiawen Li, Bohong Wu, Ya Wang, Jiao Ran, Xiao Liang, Chao Feng, et al. Vgr: Visual grounded reasoning. *arXiv preprint arXiv:2506.11991*, 2025. 1
- [48] Junjue Wang, Weihao Xuan, Heli Qi, Zhihao Liu, Kunyi Liu, Yuhan Wu, Hongruixuan Chen, Jian Song, Junshi Xia, Zhuo Zheng, et al. Disaster3: A remote sensing vision-language dataset for disaster damage assessment and response. *arXiv preprint arXiv:2505.21089*, 2025. 1, 3, 6
- [49] Weihao Xuan, Junjue Wang, Heli Qi, Zihang Chen, Zhuo Zheng, Yanfei Zhong, Junshi Xia, and Naoto Yokoya. Dynamicvl: Benchmarking multimodal large language models for dynamic city understanding. *arXiv preprint arXiv:2505.21076*, 2025. 1
- [50] Zhenghang Yuan, Zhitong Xiong, Lichao Mou, and Xiao Xi-ang Zhu. Chatearthnet: A global-scale image-text dataset empowering vision-language geo-foundation models. *Earth System Science Data Discussions*, 2024:1–24, 2024. 5, 6

- [51] Yang Zhan, Zhitong Xiong, and Yuan Yuan. Rsvg: Exploring data and models for visual grounding on remote sensing data. *TGRS*, 61:1–13, 2023. [1](#)
- [52] Yang Zhan, Zhitong Xiong, and Yuan Yuan. Skyeyegpt: Unifying remote sensing vision-language tasks via instruction tuning with large language model. *ISPRS Journal of Photogrammetry and Remote Sensing*, 221:64–77, 2025. [1](#), [2](#)
- [53] Chenhui Zhang and Sherrie Wang. Good at captioning bad at counting: Benchmarking gpt-4v on earth observation data. In *CVPR*, pages 7839–7849, 2024. [3](#)
- [54] Wei Zhang, Miaoxin Cai, Tong Zhang, Yin Zhuang, and Xuerui Mao. Earthgpt: A universal multi-modal large language model for multi-sensor image comprehension in remote sensing domain. *TGRS*, 2024. [1](#), [2](#)
- [55] Wei Zhang, Miaoxin Cai, Yaqian Ning, Tong Zhang, Yin Zhuang, He Chen, Jun Li, and Xuerui Mao. Earthgpt-x: Enabling mllms to flexibly and comprehensively understand multi-source remote sensing imagery. *arXiv preprint arXiv:2504.12795*, 2025. [2](#)
- [56] Xintong Zhang, Zhi Gao, Bofei Zhang, Pengxiang Li, Xiaowen Zhang, Yang Liu, Tao Yuan, Yuwei Wu, Yunde Jia, Song-Chun Zhu, et al. Chain-of-focus: Adaptive visual search and zooming for multimodal reasoning via rl. *arXiv preprint arXiv:2505.15436*, 2025. [2](#), [3](#)
- [57] Ziwei Zheng, Michael Yang, Jack Hong, Chenxiao Zhao, Guohai Xu, Le Yang, Chao Shen, and Xing Yu. Deep-eyes: Incentivizing” thinking with images” via reinforcement learning. *arXiv preprint arXiv:2505.14362*, 2025. [2](#), [3](#)
- [58] Yue Zhou, Mengcheng Lan, Xiang Li, Litong Feng, Yiping Ke, Xue Jiang, Qingyun Li, Xue Yang, and Wayne Zhang. Geoground: A unified large vision-language model for remote sensing visual grounding. *arXiv preprint arXiv:2411.11904*, 2024. [1](#)
- [59] Jinguo Zhu, Weiyun Wang, Zhe Chen, Zhaoyang Liu, Shenglong Ye, Lixin Gu, Hao Tian, Yuchen Duan, Weijie Su, Jie Shao, et al. Internvl3: Exploring advanced training and test-time recipes for open-source multimodal models. *arXiv preprint arXiv:2504.10479*, 2025. [2](#), [4](#), [6](#)